C Aravind

Reg no : 192512044

ECA5701-NANOELECTRONICS

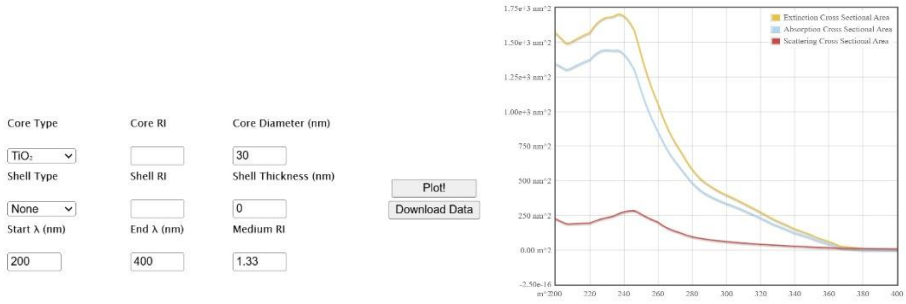
C01-1

EXPERIMENT 1

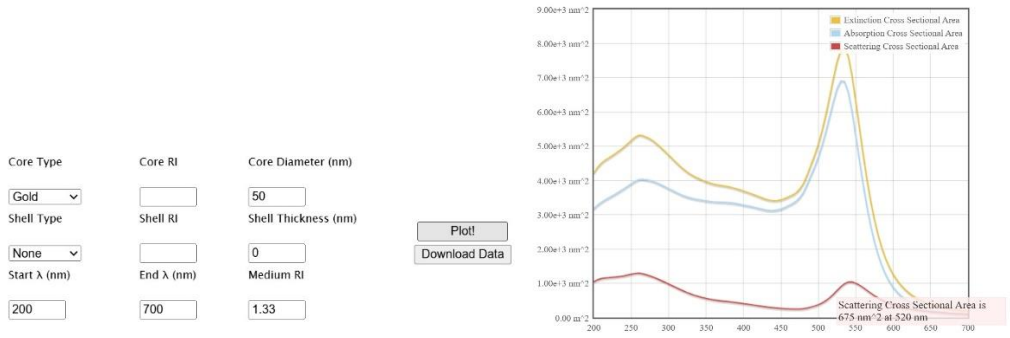
Exp. 1. Simulation of scattering and absorption of Multi-layer Sphere and CoreShell Nanoparticle using Mie-Plot tool.

Aim: To simulate and analyze the scattering, absorption, and extinction characteristics of multi-layer spheres and core–shell nanoparticles using the Mie-Plot tool.

Case Study 1: Scattering and absorption spectra for a single-layer sphere (core only).



Case Study 2: Simulate the optical response of a gold (Au) nanoparticle of 50 nm diameter in water medium.



Case Study 3: Simulate a core–shell nanoparticle with a gold core (30 nm) and silica shell (10 nm).

Core Type	Core RI	Core Diameter (nm)
Gold		30
Shell Type	**Shell RI**	**Shell Thickness (nm)**
Silica		10
Start λ (nm)	**End λ (nm)**	**Medium RI**
400	800	1.33

Plot!

Download Data

